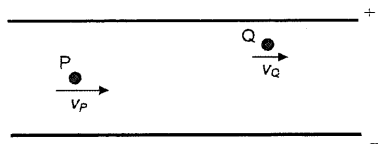


Important note: *denotes questions that may require knowledge from later chapters in order to solve them.

I. MCQs

1. [AJC/H2/Prelims 2010/P1/24]

The figure below shows two electrons P and Q, moving between two parallel charged plates with different velocities v_P and v_Q respectively.

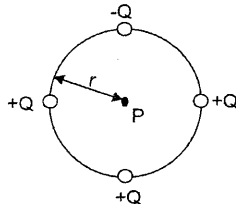


If v_P is greater than v_Q , which of the following correctly compares the potential energies and accelerations of P and Q?

	potential energy	acceleration
A	greater for P	equal for both
B	greater for P	greater for Q
C	greater for Q	equal for both
D	greater for Q	greater for P

2. [AJC/H2/Prelims 2010/P1/25]

Point P is surrounded symmetrically by four charges at distance r as shown below.



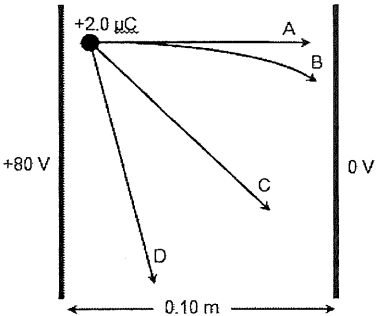
How much work is needed to move a point charge $+Q$ from infinity to point P?

- A $\frac{Q^2}{4\pi\epsilon_0 r}$
- B $\frac{Q^2}{2\pi\epsilon_0 r}$
- C $\frac{3Q^2}{4\pi\epsilon_0 r}$
- D $\frac{Q^2}{\pi\epsilon_0 r}$

3. [CJC/H2/Prelims 2010/P1/24]

A sphere of weight 1.6×10^{-3} N has an electric charge of $2.0 \mu\text{C}$. It is released from rest, in vacuum between two parallel, vertical metal plates. The separation of the plates is 0.10 m and the potential difference between them is 80 V. The point of release of the sphere is within the region of uniform electric field between the plates. The arrangement is shown in the diagram.

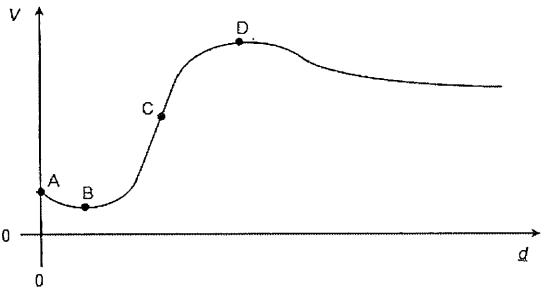
Which path does the sphere follow after release?



4. [CJC/H2/Prelims 2010/P1/25]

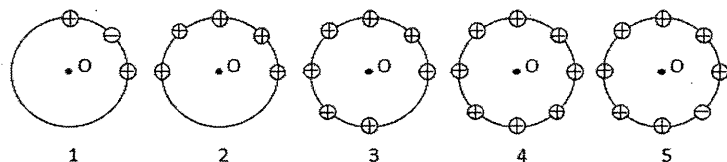
The diagram below shows the variation of the electric potential V with the distance d along a straight line in a particular electric field.

At which point is the magnitude of the electric field greatest?



5. [Dunman High/H2/Prelims 2010/P1/22]

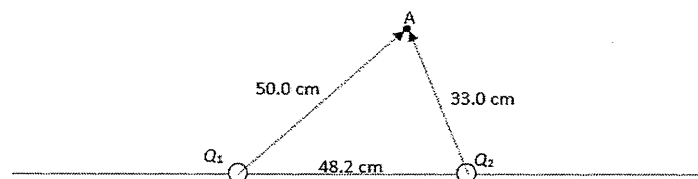
The diagrams below show 5 different arrangements of charges around a circle with centre O. Rank in *ascending* order the magnitude of the electric field strength at point O.



- A 1 2 3 5 4 B 4 1 3 5 2 C 1 2 5 3 4 D 4 3 5 2 1

6. [HCI/H2/Prelims 2010/P1/22]

Two fixed point charges, each of magnitude $Q_1 = -7.83 \text{ nC}$ and $Q_2 = +4.63 \text{ nC}$, are located at O and P respectively as shown in the figure. The distance OP is 48.2 cm.

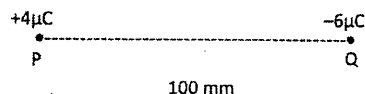


What is the value of the electric potential at point A?

- A -14.7 V B $+29.7 \text{ V}$ C $+101 \text{ V}$ D -267 V

7. [MI/H2/Prelims 2010/P1/24]

Two charges, P and Q are 100 mm apart.



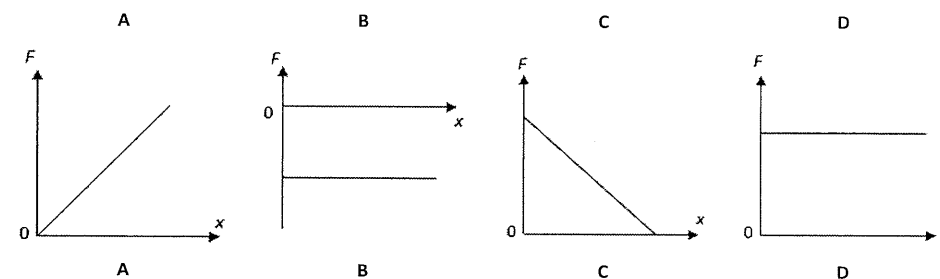
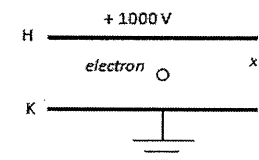
X is a point on the line between P and Q. If the potential at X is 0 V, what is the distance from P to X?

- A 40 mm B 45 mm C 50 mm D 60 mm

8. [HCI/H2/Prelims 2010/P1/24]

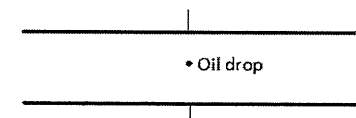
An electron is located midway inside the space between horizontal metal plates H and K. A potential difference of 1000 V is applied across H and K.

H is positively charged while K is earthed. Which one of the following graphs best represents how the electric force F acting on the electron varies with the distance x measured from plate H?



9. [MJC/H2/Prelims 2010/P1/23]

A negatively-charged oil drop is held stationary between two horizontal, charged metal plates, the upper plate being positive.



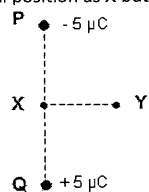
The oil-drop then acquires an additional negative charge.

In order to keep the oil-drop stationary, what change should be made?

- A Move the plate closer together.
B Reverse the charges on the plates.
C Increase the electric field strength between the plates.
D Decrease the potential difference between the plates.

10. [MJC/H2/Prelims 2010/P1/24]

Two point charges of $-5\ \mu\text{C}$ and $+5\ \mu\text{C}$, are situated at points P and Q respectively as shown below. X lies midway between P and Q while Y is at the same vertical position as X but displaced to the right.

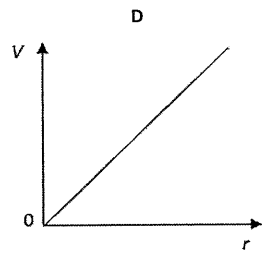
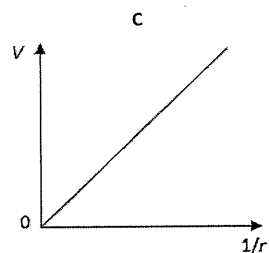
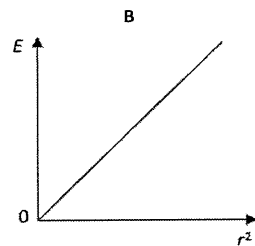
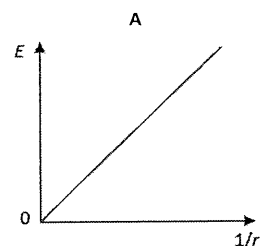


Which of the following is true for point X and point Y?

	At point X		At point Y	
	electric field	electric potential	electric field	electric potential
A	towards Q	zero	downwards	zero
B	towards Q	negative	downwards	negative
C	towards P	zero	upwards	zero
D	towards P	negative	upwards	negative

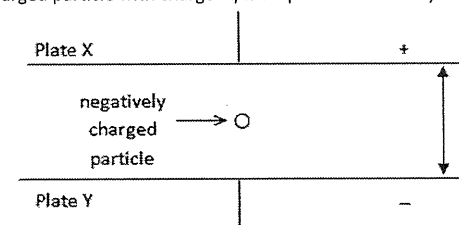
11. [NJC/H2/Prelims 2010/P1/25]

Which graph correctly relates the electric field strength E or electric potential V in the field of a positive point charge, with a distance r from the charge?



12. [NJC/H2/Prelims 2010/P1/26]

Two large parallel metal plates X and Y are situated in a vacuum as shown. The potential difference between plate X and Y is V . A negatively charged particle with charge $-q$ is suspended stationary midway between the 2 plates.



Which statement is **not** correct?

- A When V is halved, the particle will accelerate downwards with an acceleration of $0.5\ g$.
- B When V is halved, the particle will reach plate X with kinetic energy equals $0.5\ qV$.
- C When V is doubled, the particle will accelerate upwards with an acceleration of g .
- D When V is doubled, the particle will reach plate X with kinetic energy equals $0.5\ qV$.

13. [SAJC/H2/Prelims 2010/P1/27]

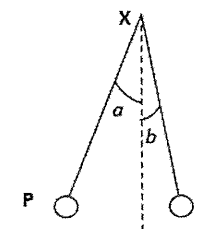
The force of attraction between two unlike charges is $1.5\ \text{N}$.

If the distance between the charges is doubled, the force of attraction between them is

- A $0.75\ \text{N}$
- B $0.38\ \text{N}$
- C $3.0\ \text{N}$
- D $6.0\ \text{N}$

14. [NYJC/H2/Prelims 2010/P1/23]

The diagram below shows two small charged spheres P and Q of small mass which are hung by identical fine nylon threads from a fixed point X. It is found that, in equilibrium, the angle a is greater than the angle b .



Which statement is correct?

- A The mass of P is less than that of Q.
- B The mass of P is greater than that of Q.
- C The charge of P is numerically smaller than that of Q.
- D The charge of P is numerically greater than that of Q.

15. [NYJC/H2/Prelims 2010/P1/24]

A metal sphere of radius 0.1 m was insulated from its surroundings and given a large positive charge. A small charge was brought from a distant point to a point 0.5 m from the sphere's centre. The work done against the electric field was W and the force on the small charge in its final position was F .

If the small charge had been moved to only 1 m from the centre of the sphere, what would have been the values for the work done and the force?

	work done	force
A	$W/4$	$F/2$
B	$W/2$	$F/4$
C	$W/2$	$F/2$
D	$W/2$	$F/\sqrt{2}$

16. [PJC/H2/Prelims 2010/P1/24]

The electric potentials V are measured at distances x from P along a line PQ.

The results are:

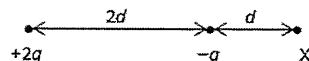
V / V	13	15	18	21	23
x / m	0.020	0.030	0.040	0.050	0.060

The component along PQ of the electric field for $x = 0.040 \text{ m}$ is approximately

- A 300 V m^{-1} towards Q.
 B 300 V m^{-1} towards P.
 C 450 V m^{-1} towards Q.
 D 450 V m^{-1} towards P.

17. [RI/H2/Prelims 2010/P1/23]

Two charges $+2q$ and $-q$ are placed at a distance $2d$ apart.



What is the electric potential at X, a distance d away from $-q$?

- A $\frac{-q}{4\pi\epsilon_0 d}$ B $\frac{-q}{12\pi\epsilon_0 d}$ C $\frac{-q}{36\pi\epsilon_0 d^2}$ D $\frac{-7q}{36\pi\epsilon_0 d^2}$

18. [RVHS/H2/Prelims 2010/P1/22]

X and Y are two identical conducting spheres separated by a distance d . X has a charge of $+6 \mu\text{C}$ and Y has a charge of $-2 \mu\text{C}$. The electric force between them is $-F$ (i.e. attractive). The spheres are touched together and are then returned to their original separation d .

The force between them now is nearest to

- A $+F$ B $-F$ C $+\frac{F}{3}$ D $-\frac{F}{3}$

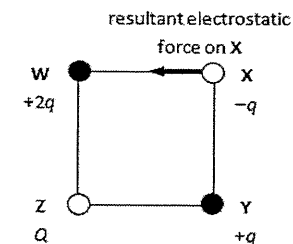
19. [SAJC/H2/Prelims 2010/P1/25]

Which statement about the electric potential and electric field at a point is correct?

- A The potential at the point is always zero when the electric field at that point is zero.
 B The electric potential is given by the rate of change of electric field intensity with distance.
 C The electric field at a point is zero when the potential around the point is constant.
 D The potential at a point is zero when the electric field around the point is constant.

20. [SAJC/H2/Prelims 2010/P1/8]

Three charges $+2q$, $-q$ and $+q$ are placed at the corners W, X and Y of a square WXYZ respectively as shown.

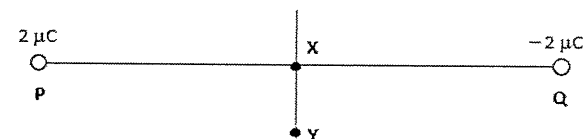


A fourth charge Q is placed at Z, after which the charge X experiences a net electrostatic force indicated by the arrow in the above diagram. What is the value of Q ?

- A $-2.8q$ B $-1.4q$ C $1.4q$ D $2.8q$

21. [SAJC/H2/Prelims 2010/P1/28]

Charges of $2 \mu\text{C}$ and $-2 \mu\text{C}$ situated at points P and Q respectively as shown. X lies midway between P and Q while Y is at a mid-line.

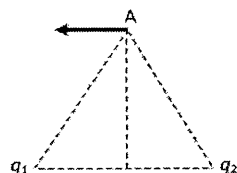


Which of the following correctly describes the directions of the electric fields and electric potentials at points X and Y?

	At point X		At point Y	
	electric Field	potential	electric Field	potential
A	right	zero	right	zero
B	upwards	positive	upwards	negative
C	right	zero	left	zero
D	downwards	negative	downwards	positive

22. [TJC/H2/Prelims 2010/P1/25]

Two charges q_1 and q_2 are placed at two corners of an equilateral triangle as shown. The resultant electric field at point A is directed to the left as indicated.



How is charge q_2 related to charge q_1 ?

- A $q_2 = q_1$ B $q_2 = \sqrt{2}q_1$ C $q_2 = -q_1$ D $q_2 = -\sqrt{2}q_1$

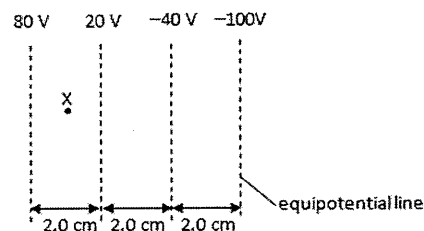
23. [VJC/H2/Prelims 2010/P1/22]

Which statement about an electric field is **not** correct?

- A The electric field strength due to a point charge varies as $1/r^2$ where r is the distance from the charge.
 B Electric field strength is a vector quantity.
 C The electric field strength at a point is a measure of the potential gradient at that point.
 D The electric field strength is zero at all points where the potential is zero.

24. [VJC/H2/Prelims 2010/P1/23]

The diagram shows equipotential lines spaced 2.0 cm apart.



What is the electric force which is exerted on a charge of $+5.0 \mu\text{C}$ when placed at X as shown?

- A $3.0 \times 10^{-2} \text{ N}$ to the right
 B $3.0 \times 10^{-2} \text{ N}$ to the left
 C $1.5 \times 10^{-2} \text{ N}$ to the right
 D $1.5 \times 10^{-2} \text{ N}$ to the left

25. [YJC/H2/Prelims 2010/P1/21]

Which statement about an electric field is **not** correct?

- A The electric field strength due to a point charge varies proportionately with $1/r^2$ where r is the distance from the charge.
 B The electric field strength at a point is a measure of the force exerted on a unit positive charge at that point.
 C The negative sign in the formula $E = -\frac{dV}{dr}$, implies that the field points in the direction of decreasing potential.
 D The electric field strength at a point is equal to the potential gradient at that point.

II. Structured Questions

1. [AJC/H2/Prelims 2010/P3/2 (part)]

Two charged points A and B are separated by a distance of 30.0 cm, as shown in Fig. 1.1.

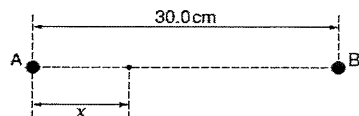


Fig. 1.1

The variation with distance x from A along AB of the potential V is shown in Fig. 1.2.

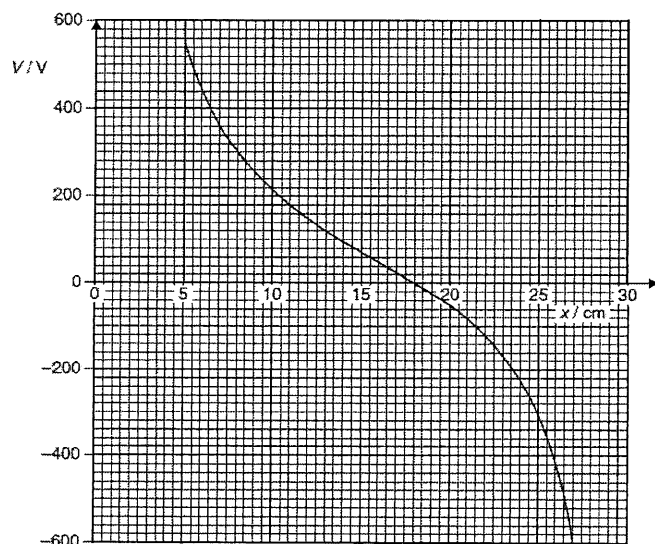


Fig. 1.2

A test charge is moved along the line AB from $x = 5.0$ cm to $x = 27$ cm.

- Describe and explain the variation with distance x of the force acting on the test charge. [3]
- State the value of x at which the force on the test charge is maximum. [1]

[Total: 4]

2. [Dunman High/H2/Prelims 2010/P3/6]

Ink-jet printing is a non-impact, dot matrix technology in which ink droplets are projected through an array of tiny nozzles onto paper. In a continuous flow print head, ink-jet droplets are approximately $80 \mu\text{m}$ in diameter and can be charged negatively as they pass through a charging ring electrode immediately after they break away from the nozzle of the print head. Whether or not a droplet is charged is controlled by a signal sent to the printer by a computer. An ink droplet that becomes charged would be deflected into a gutter when it passes through a region with a uniform electric field strength while uncharged droplets pass through undeflected to reach the paper.

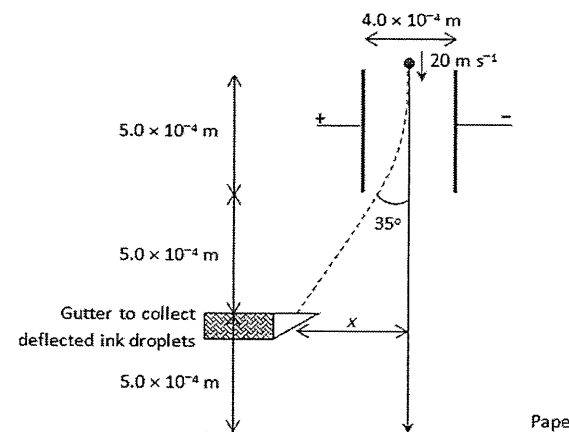


Fig. 2.1

- Fig. 2.1 shows the parallel deflecting plates with some dimensions of the ink-jet cartridge. The ink drops fall straight through the centerline between the deflecting plates with a constant vertical velocity of 20 m s^{-1} . In order to land in the center of the gutter, a charged ink droplet must leave the deflecting plates at an angle of 35° .
 - Calculate the horizontal component of velocity as a charged ink droplet leaves the deflecting plates. [1]
 - Determine the horizontal acceleration of the charged ink droplet for which it is between the deflecting plates. [3]
 - Determine the horizontal distance x between the centre of the gutter and the centerline of between the deflecting plates. [3]
- Explain the term *uniform electric field strength*. [2]
 - On Fig. 2.1, sketch the electric lines representing electric field in the region between the pair of parallel deflecting plates. [2]
 - Each ink droplet has a mass of $2.9 \times 10^{-10} \text{ kg}$. Using your answer to a.ii., determine the electric force F_E acting on the charged ink droplet while it is between the deflecting plates. [2]
 - If the charge gained by the ink droplet is $-2 \times 10^{-10} \text{ C}$, determine the electric field strength E between the deflecting plates. [2]
 - Hence, calculate the potential difference V between the deflecting plates. [2]
- The uncharged, undeflected ink droplets travel beyond the deflecting plates towards the paper. When a droplet hits the page, it forms a dot sticking to it. Assuming that the printer giving print of 20 dots per mm would take 0.08 s to print a line on a page 20.0 cm wide, calculate the average force exerted on the page when the ink droplets strike. The ink has a density of 1100 kg m^{-3} . [3]

[Total: 20]

3. [MJC/H2/Prelims 2010/P2/2 (part)]

Air near the Earth conducts electricity to a small extent. Because there is an electric field directed towards the Earth's surface, a small electric current exists in the atmosphere. Measurements of the electric field near the Earth's surface give an average of about $1.0 \times 10^2 \text{ V m}^{-1}$.

- Estimate the potential difference between the head and feet of an average adult. [1]
- The electric field near the Earth's surface is as large as $1.0 \times 10^2 \text{ V m}^{-1}$.
Explain why you do not experience an electric shock when you stand upright. [2]

[Total: 3]

4. [NJC/H2/Prelims 2010/P2/3]

Fig. 4.1 shows three negative point charges ($-q$, $-q$, $-2q$) and one positive point charge ($+4q$), along with some electric field lines drawn between the charges. The 3 negative charges are at the same distance from the positive charge.

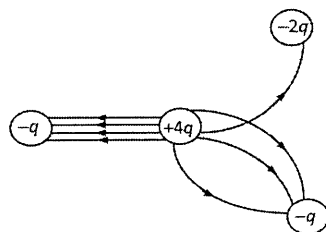


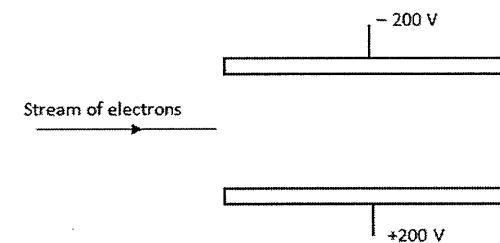
Fig. 4.1

- Identify and explain any 2 of the mistakes in this figure. [2]
- On a separate diagram, draw the correct distribution of electric field lines. [2]
- If the distance from the $+4q$ charge in Fig.4.1 to all the other 3 charges is x , determine in terms of q and x the work done by the external agent in order to bring the $+4q$ charge to infinity. [2]

[Total: 6]

5. [NYJC/H2/Prelims 2010/P2/5]

- Define *electric field strength*. [1]
- Fig. 5.1 shows a stream of electrons entering a region between two parallel plates which have a potential difference.



- On Fig. 5.1, draw the electric field lines between the plates and the expected path of the stream of electrons between and after the plates. [2]
- Calculate the distance between the plates given that the electric field strength between the plates is $2.0 \times 10^4 \text{ N C}^{-1}$. [1]
- Calculate the acceleration of the electrons between the plates. [2]
- Hence, given that the length of each plate is 0.040 m and initial horizontal speed of the electrons is $1.0 \times 10^6 \text{ m s}^{-1}$, calculate the vertical deflection of the electron at the end of the plates. [2]

[Total: 7]

6. [RI/H2/Prelims 2010/P2/5]

Two small identical styrofoam balls of mass 0.50 g and charge $+15 \text{ nC}$ are placed in a hemispherical bowl of radius R with frictionless, non-conducting walls. At equilibrium, they are at a distance of $0.50R$ apart as shown in Fig. 6.1.

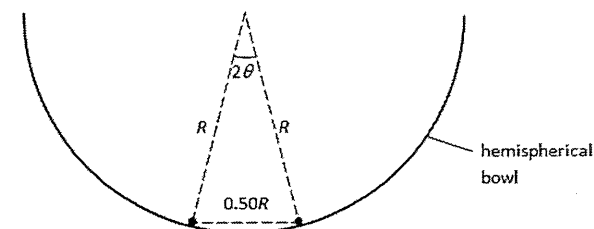


Fig. 6.1

- Show that $\theta \approx 14.5^\circ$. [1]
 - On Fig. 6.1, indicate all the forces acting on **one** of the balls in the bowl. [2]
 - Hence, determine the radius R of the bowl. [3]
- State the effect on θ if the Styrofoam balls are replaced with metal ones of the same size and charge. [1]

[Total: 7]

7. [RVHS/H2/Prelims 2010/P2/4]

- a. A and B are two identical conducting spheres each carrying a charge $+Q$. They are placed in a vacuum with their centres distance d apart as shown in Fig. 7.1

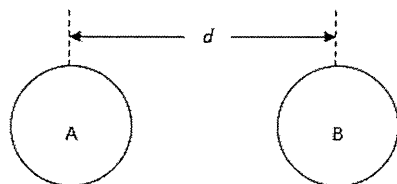


Fig. 7.1

Explain why the force F between them is **not** given by the expression $F = \frac{Q^2}{4\pi\epsilon_0 d^2}$.

[2]

- b. Electric fields and magnetic fields may be represented by lines of force. Fig. 7.2 shows some lines of force.

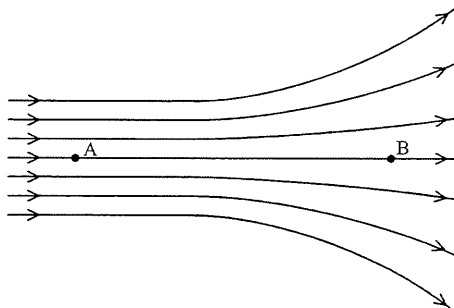


Fig. 7.2

- State whether the field strength at the vicinity of A and at the vicinity of B is constant, increasing or decreasing when measured in the direction from A towards B.
- Explain why the field lines can never touch or cross.

[2]

[2]

[Total: 6]

8. [TJC/H2/Prelims 2010/P3/2]

- a. Fig. 8.1 shows some of the electric field lines around a metal needle which is held at a high positive voltage.

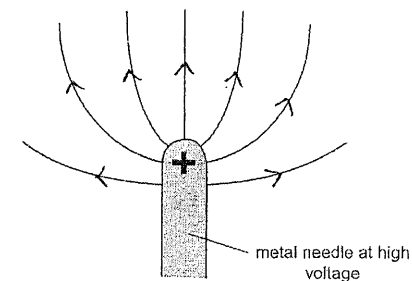


Fig. 8.1

- Draw two equipotential lines around the charged metal needle.
 - Suggest the effect of the electric field on a single air molecule near the needle tip as electrical breakdown of the air occurs.
- b. Fig. 8.2 shows a charged conducting sphere of diameter 4.0 mm which is connected to an insulating rod. The electric field strength at the surface of the conducting sphere is $1.13 \times 10^7 \text{ V m}^{-1}$.

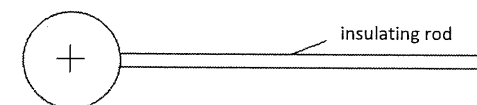


Fig. 8.2

- Calculate
 - the magnitude of the charge on the conducting sphere,
 - the electric potential of the charged sphere.
- The charged sphere is attached to a top-pan balance as shown in Fig. 8.3. A second identically charged sphere is then clamped vertically above this sphere such that their centres are 4.0 cm apart.

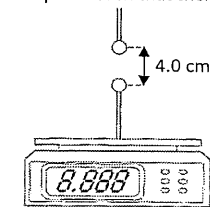


Fig. 8.3

The balance can record masses to the nearest 0.001 g. The initial reading on the balance before the second charged sphere is clamped above the first sphere is 8.205 g.

- Assuming that the spheres behave like point charges, determine the final reading on the balance.
- State and explain whether your answer to b.ii.1. is an under-estimate or an over-estimate if the assumption that the spheres behave like point charges is not valid.

[Total: 9]

9. [VJC/H2/Prelims 2010/P2/1]

- a. Define *electric potential energy* at a point. [1]
- b. Fig. 9.1 below shows the path of a proton (${}^1_1\text{p}$) travelling head-on towards a uranium (${}^{235}_{92}\text{U}$) nucleus.

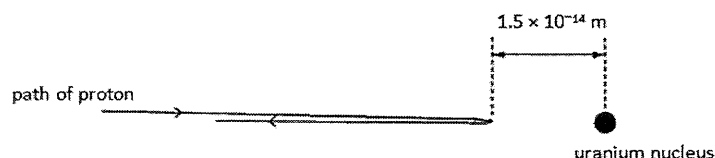


Fig. 9.1

By considering only the electrostatic repulsion, the distance of closest approach of the proton to the uranium nucleus is $1.5 \times 10^{-14} \text{ m}$.

- State the form of the energy to which the kinetic energy of the proton has been converted when it is at the distance of closest approach. [1]
- Calculate the initial speed of the proton and state the assumption made in the calculation. [4]
- In nuclear fission reactions, protons are not normally used to induce the reactions. Suggest an alternative particle that you would use to penetrate the nucleus and explain your choice. [2]

[Total: 8]

10. [YJC/H2/Prelims 2010/P3/7]

- Define *electric field strength* and state the SI unit in which it is measured. [2]
- Two charged parallel plates are separated by a distance d and have a potential difference V between them.
Write down an expression for the electric field strength of the uniform field between the plates. [1]

- b. Fig. 10.1 illustrates two plates A and B, at a distance 30 mm apart in a vacuum, with plate A at a potential of -4.2 V and plate B at a potential of -2.0 V .

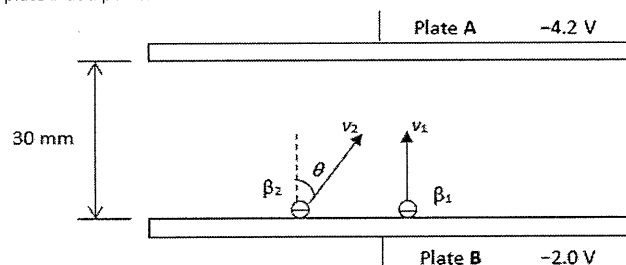


Fig. 10.1

- Draw arrows to represent the electric field between the plates. [1]
- Calculate the magnitude of this electric field strength. [2]
- An electron, β_1 , is emitted from plate B in a direction normal to its surface and moves directly towards plate A. Calculate the minimum velocity, v_1 with which the electron needs to be emitted in order to reach plate A. [2]

- iv. State, with a reason, how your answer in b.iii. may be affected if

- the distance between the plates had been halved to 15 mm, while keeping the potential difference the same, [2]
- the potential of plate A is changed to $+4.2 \text{ V}$ and potential of plate B is changed to $+2.0 \text{ V}$ while keeping the distance between the plates at 30 mm. [2]

- v. Another electron, β_2 , is emitted from plate B with a velocity $v_2 = 1.1 \times 10^6 \text{ m s}^{-1}$ at an angle θ to the normal as shown in Fig. 10.1. Determine the largest possible angle θ which would allow the electron to just reach plate A. [2]

- c. The potentials of plates A and B are now changed to $+0.29 \text{ V}$ and 0 V respectively. A charged micro-particle is suspended between the two plates as shown in Fig. 10.2.

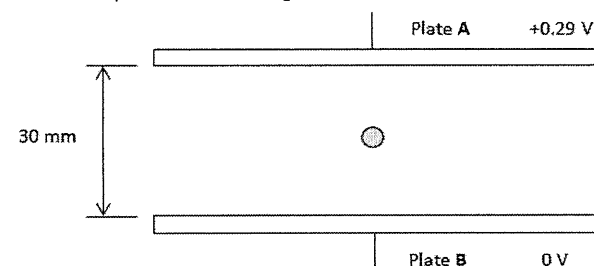


Fig. 10.2

- State the sign of charge present on the particle. [1]
- Determine the charge to mass ratio, q/m of the particle. [2]
- A student claims that the mass of the particle is $72 \times 10^{-19} \text{ kg}$.
Based on your answer to part c.ii., explain why his answer is not valid. [2]

[Total: 20]

III. Solutions and Answers

The suggested solutions and answers below are only for your reference and have not been verified to be error-free. If you have any doubt, please discuss with your tutor to clarify.

MCQs

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A	B	C	C	B	A	A	B	D	C	C	B	B	A	B

16	17	18	19	20	21	22	23	24	25
B	B	C	C	A	A	C	D	C	D

- 1 A Electric field strength E between the plates is constant.
 Since P is at a lower potential (nearer lower potential plate) and is negatively charged, its electric potential energy is higher. ($U = qV = -eV$)

$$\text{acceleration, } a = \frac{F}{m} = \frac{Eq}{m} = \frac{Ee}{m}$$

Since E is constant, the accelerations of P and Q are the same.

- 2 B Electric potential at P, $V_P = \frac{-Q}{4\pi\epsilon_0 r} + 3\left(\frac{Q}{4\pi\epsilon_0 r}\right) = \frac{2Q}{4\pi\epsilon_0 r}$

Work done causes a change in the potential energy of the system when another point charge is moved from infinity to P.

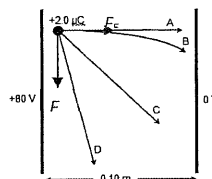
$$W = \Delta U = Q(V_P - 0) = Q\left(\frac{2Q}{4\pi\epsilon_0 r}\right) = \frac{Q^2}{2\pi\epsilon_0 r}$$

- 3 C $F_G = 1.6 \times 10^{-3} \text{ N}$

$$F_E = Eq = \frac{\Delta V}{d} \cdot q = \frac{80}{0.10} (2.0 \times 10^{-6}) = 1.6 \times 10^{-3} \text{ N}$$

Since $|F_G| = |F_E|$ and their magnitudes remain constant, the resultant force acting on the sphere is also constant and acting at the middle between F_G and F_E .

The path of the sphere will follow the direction of the constant resultant force.



- 4 C $|E| = \left| -\frac{dV}{dr} \right|$

The magnitude of the electric field strength is given by the magnitude of the gradient of the graph. At point C, magnitude the gradient is the greatest.

- 5 B Electric field strength E is a vector quantity.

$$\text{magnitude of electric field strength given by } E = \frac{Q}{4\pi\epsilon_0 r^2}$$

direction of E is in the direction of the force acting on a positive charge

resultant field strength at point O is found from the vector addition of the individual field strength due to each charge

determine the resultant field strength at point O for each of the charge-arrangement and compare

- 6 A Electric potential is a scalar quantity. The electric potential at A is found from the algebraic summation of the potentials due to Q_1 and Q_2 .

$$\begin{aligned} V_A &= V_{Q_1} + V_{Q_2} \\ &= \frac{Q_1}{4\pi\epsilon_0 r_1} + \frac{Q_2}{4\pi\epsilon_0 r_2} \\ &= \frac{1}{4\pi(8.85 \times 10^{-12})} \left(\frac{-7.83 \times 10^{-9}}{0.500} + \frac{4.63 \times 10^{-9}}{0.330} \right) \\ &= -14.7 \text{ V} \end{aligned}$$

- 7 A Let d be the distance between P and X.

$$\begin{aligned} V_X &= \frac{Q_P}{4\pi\epsilon_0 d} + \frac{Q_Q}{4\pi\epsilon_0 (0.100 - d)} \\ 0 &= \frac{Q_P}{4\pi\epsilon_0 d} + \frac{Q_Q}{4\pi\epsilon_0 (0.100 - d)} \\ d &= \frac{4(0.100)}{10} = 40 \text{ mm} \end{aligned}$$

- 8 B Since the electric field strength E between the plates is uniform, the electric force F acting on the electron is constant as $F = Ee$.

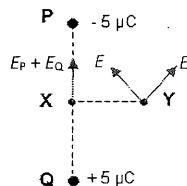
The direction of F on the negatively charged electron is upwards towards the higher potential plate, which is opposite to the positive downwards direction of x as indicated on the diagram. Hence the direction of F is negative.

- 9 D Forces acting on the oil drop: electric force F_E acting upwards and weight W acting downwards
 When the oil drop is stationary,

$$F_E = W \Rightarrow Eq = W \Rightarrow \frac{\Delta V}{d} q = W$$

Since W is constant, when the value of q increases, ΔV needs to decrease or d needs to increase or E needs to decrease.

10 C $E = \frac{Q}{4\pi\epsilon_0 r^2}$ (vector)
 $V = \frac{Q}{4\pi\epsilon_0 r}$ (scalar)



At X, electric field strength due to charge at P is vertically upwards towards P and electric field strength due to charge at Q is also vertically upwards towards P. Hence the resultant field strength is also vertically upwards towards P.

At X, electric potential due to charge at P is negative and electric potential due to charge at Q is positive. They are of the same magnitude. Hence the net electric potential is zero.

At Y, the direction of the electric field strength due to each charge is as shown above. The magnitudes of the electric field strengths are the same. Hence the resultant field strength is vertically upwards at Y.

At Y, At X, electric potential due to charge at P is negative and electric potential due to charge at Q is positive. They are of the same magnitude. Hence the net electric potential is zero.

11 C $E = \frac{Q}{4\pi\epsilon_0 r^2}$, $V = \frac{Q}{4\pi\epsilon_0 r}$

$$E \propto \frac{1}{r^2}, V \propto \frac{1}{r}$$

Option C shows the correct proportionality.

- 12 B Forces acting on the oil drop: electric force F_E acting upwards and weight W acting downwards
 When the oil drop is stationary,

$$F_E = W \Rightarrow Eq = W \Rightarrow \frac{V}{d}q = W = mg$$

A change to V will result in a resultant force acting on the charged particle:

$$\frac{V'}{d}q - mg = ma$$

Option B is not correct: When $V' = \frac{1}{2}V$, the upwards electric force will decrease and the resultant force on the particle is downwards. Hence the particle will accelerate downwards and reach plate Y, not X.

13 B $F_E = \frac{Q_1 Q_2}{4\pi\epsilon_0 r^2} = 1.5 \text{ N}$

When distance is doubled,

$$F_E' = \frac{Q_1 Q_2}{4\pi\epsilon_0 (2r)^2} = \frac{Q_1 Q_2}{4\pi\epsilon_0 (4r^2)} = \frac{1}{4} \left(\frac{Q_1 Q_2}{4\pi\epsilon_0 r^2} \right) = \frac{1}{4} (1.5) = 0.38 \text{ N}$$

- 14 A Sphere P:
 $T_P \cos a = m_P g$
 $T_P \sin a = F_E$

- Sphere Q:
 $T_Q \cos b = m_Q g$
 $T_Q \sin b = F_E$

$$T_P \sin a = T_Q \sin b$$

$$\text{Since } a > b \Rightarrow \sin a > \sin b \Rightarrow T_P < T_Q$$

$$\text{Hence } T_P \cos a < T_Q \cos b \text{ as } \cos a < \cos b$$

$$m_P g < m_Q g \Rightarrow m_P < m_Q$$

15 B $W_0 = U = \frac{Q_1 Q_2}{4\pi\epsilon_0 r} \Rightarrow W_0 \propto \frac{1}{r}$

$$\frac{W_2}{W_1} = \frac{r_1}{r_2} \Rightarrow W_2 = \frac{0.5}{1} (W) = \frac{W}{2}$$

$$F_0 = \frac{Q_1 Q_2}{4\pi\epsilon_0 r^2} \Rightarrow F_0 \propto \frac{1}{r^2}$$

$$\frac{F_2}{F_1} = \frac{r_1^2}{r_2^2} \Rightarrow F_2 = \frac{0.5^2}{1^2} (F) = \frac{F}{4}$$

16 B $E = -\frac{dV}{dx} \approx -\frac{\Delta V}{\Delta x}$
 $|E| = \frac{\Delta V}{\Delta x} = \frac{21-15}{0.050-0.030} = 300 \text{ V m}^{-1}$

E always points in the direction of decreasing potential i.e. towards P.

- 17 B Electric potential is a scalar quantity.

$$V_x = \frac{2q}{4\pi\epsilon_0 (3d)} + \left(\frac{-q}{4\pi\epsilon_0 (d)} \right) = \frac{q}{4\pi\epsilon_0 d} \left(\frac{2}{3} - 1 \right) = -\frac{q}{12\pi\epsilon_0 d}$$

- 18 C Before spheres are touched together,

$$F_E = \frac{Q_1 Q_2}{4\pi\epsilon_0 r^2} = \frac{(+6)(-2)\mu}{4\pi\epsilon_0 r^2} = \frac{-12\mu}{4\pi\epsilon_0 r^2} = -F$$

After spheres are touched, charges will distribute themselves evenly across both and each will have charge of $+2\mu\text{C}$.

$$F_E' = \frac{Q_1 Q_2}{4\pi\epsilon_0 r^2} = \frac{(+2)(+2)\mu}{4\pi\epsilon_0 r^2} = \left(-\frac{4}{12} \right) \left(\frac{-12\mu}{4\pi\epsilon_0 r^2} \right) = \left(-\frac{1}{3} \right) (-F) = \frac{1}{3} F$$

- 19 C Electric field points from a region of higher potential to a region of lower potential. There is no electric field if there is no potential difference i.e. potential is constant.

$$E = -\frac{dV}{dr}$$

If $E = 0$, $\frac{dV}{dr} = 0$ i.e. potential does not change with distance and is hence constant

- 20 A Since the resultant force on X is horizontal when charge at Z is present, the vertical component of the force on X by Z must be equal and opposite to the force on X by Y i.e. upwards. And since the force on X by Z is along the diagonal XZ, the force needs to point away from Z. Hence, the charge Q at Z is negatively charged.

Let the length of each side of the square be d .

$$\text{magnitude of the vertical component of force on X by Z, } F_v = \frac{q^2}{4\pi\epsilon_0 d^2}$$

$$\text{magnitude of the force on X by Z, } F = \frac{|Q|q}{4\pi\epsilon_0 (\sqrt{2}d)^2} = \frac{|Q|q}{4\pi\epsilon_0 (2d^2)}$$

Since the force on X by Z is along the diagonal, it is at an angle of 45° with the horizontal.

This means $F_v = F_h$ where F_h is the magnitude of the horizontal component of F (isosceles right angle triangle).

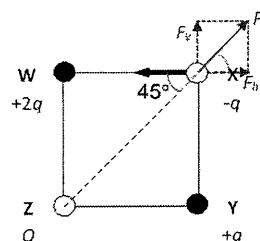
By Pythagoras Theorem,

$$F = \sqrt{F_h^2 + F_v^2} = \sqrt{2F_v^2} = \sqrt{2}F_v$$

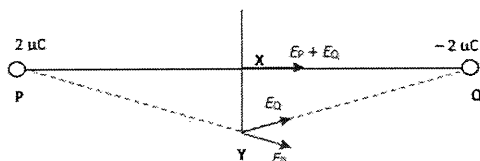
$$\frac{|Q|q}{4\pi\epsilon_0 (2d^2)} = \sqrt{2} \frac{q^2}{4\pi\epsilon_0 d^2}$$

$$|Q| = 2\sqrt{2}q = 2.8q$$

$$Q = -2.8q$$



21 A



$$E = \frac{Q}{4\pi\epsilon_0 r^2}, V = \frac{Q}{4\pi\epsilon_0 r}$$

Electric field strength is a vector quantity and the resultant at a point is found from the vector addition of the individual electric field strength at that point.

Electric potential is a scalar quantity and the net potential at a point is the algebraic summation of the individual potential at that point.

At point X:

As seen from the diagram above, the resultant electric field strength to the right.

The potential due to the positive charge is positive and the potential due to the negative charge is negative.

The 2 potentials have the same magnitude. Hence the net potential is zero.

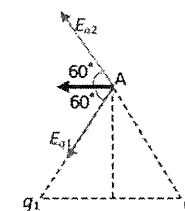
At point Y:

As seen from the diagram above, the resultant electric field strength is horizontally to the right as E_P and E_Q have the same magnitudes.

The 2 potentials have the same magnitude but different signs. Hence the net potential is zero.

- 22 C Since the resultant electric field is horizontally to the left, it means that the vertical components of the electric field strength due to q_1 and q_2 are equal in magnitude and opposite in direction such that the resultant is zero in the vertical direction.

The horizontal components of the electric field strength due to q_1 and q_2 must also be such that their resultant is to the left.



To satisfy these conditions, the directions of the electric field strengths, E_{q1} and E_{q2} , due to q_1 and q_2 respectively, can only be as that drawn in the diagram above.

$$\text{And } |E_{q1}| = |E_{q2}| \Rightarrow \left| \frac{q_1}{4\pi\epsilon_0 r} \right| = \left| \frac{q_2}{4\pi\epsilon_0 r} \right| \Rightarrow |q_1| = |q_2|$$

Since E points in the direction of the force acting on a small positive test charge, q_1 is negatively charged and q_2 is positively charged. Hence, $-q_1 = q_2$.

- 23 D Where electric field strength is zero, the potential is the same everywhere. Potential does not need to be zero.

$$E = -\frac{dV}{dr}$$

If $E = 0$, $\frac{dV}{dr} = 0$ i.e. potential does not change with distance and is hence constant but V need not be zero.

- 24 C Field is uniform since for every 2.0 cm, potential difference is consistently 60 V.

$$\text{uniform field strength, } |E| = \frac{\Delta V}{d} = \frac{60}{2.0 \times 10^{-2}}$$

$$|F_x| = |E_x|q = \frac{60}{2.0 \times 10^{-2}} (5.0 \times 10^{-6}) = 1.5 \times 10^{-2} \text{ N}$$

The direction of the electric force on the positive charge at X will be towards the lower potential region i.e. to the right.

- 25 D

$$E = -\frac{dV}{dr}$$

The electric field strength at a point is equal to the *negative* of the potential gradient at that point.

Structured Questions

1. [AJC/H2/Prelims 2010/P3/2 (part)]

- a. Force decreases in magnitude then increases in magnitude. (Force is minimum in magnitude at $x = 18$ cm (least steep gradient) and maximum in magnitude at $x = 27$ cm (steepest gradient))

Field strength = $-\text{gradient of graph}$

Force = charge $\times$ field strength OR force $\propto$ field strength

- b. $x = 27$ cm

2. [Dunman High/H2/Prelims 2010/P3/6]

- a. i. $\tan 35^\circ = \frac{v_x}{v_y} \Rightarrow v_x = v_y \tan 35^\circ = (20) \tan 35^\circ = 14.0 \text{ m s}^{-1}$

- ii. Consider the vertical motion, time taken for the droplet to pass through the deflecting plates:

$$t = \frac{5.0 \times 10^{-4}}{20} = 2.50 \times 10^{-5} \text{ s}$$

Consider the horizontal motion, the horizontal acceleration the droplet experiences is:

$$a_x = \frac{v_x - u_x}{t} = \frac{14.00 - 0}{2.5 \times 10^{-5}} = 5.60 \times 10^5 \text{ m s}^{-2}$$

- iii. Distance from centreline when charged ink droplet leaves the plates

$$x_1 = u_x t + \frac{1}{2} a_x t^2 = 0 + \frac{1}{2} (5.6 \times 10^5) (2.5 \times 10^{-5})^2 = 1.75 \times 10^{-4} \text{ m}$$

Thus, the horizontal distance x between the centre of the gutter and the centerline of between the deflecting plates:

$$x = x_1 + x_2 = 1.75 \times 10^{-4} + (5.0 \times 10^{-4}) \tan 35^\circ = 5.25 \times 10^{-4} \text{ m}$$

- b. i. Electric field strength at a point is the electric force exerted per unit positive charge placed at that point in the field.

Uniform means that every point in the field region has the same (not constant) field strength.

Reason why 'constant' is not accepted: the electric field may be time-varying, and so not constant. However, 'uniform' means that no matter how the electric field varies, every point within the concerned region always has the same electric field.

- ii. Left to right; Equally spaced (minimum 3 field lines)

- iii. $F_e = m_{\text{ink}} a_y = (2.9 \times 10^{-10}) (5.6 \times 10^5) = 1.62 \times 10^{-4} \text{ N}$

- iv. $E = \frac{F_e}{q} = \frac{1.624 \times 10^{-4}}{2 \times 10^{-10}} = 8.12 \times 10^5 \text{ V m}^{-1}$

- v. $E = \frac{\Delta V}{d} \Rightarrow \Delta V = Ed = (8.12 \times 10^5) \times (4.0 \times 10^{-4}) = 325 \text{ V}$

- c. Total number of ink drops required: $(20) \times (200) = 4000$

$$\text{By Newton's Second Law: } F_y = \frac{\Delta p_y}{t} = \frac{M \Delta v_y}{t} = \frac{4000 \times (2.95 \times 10^{-10}) \times (20 - 0)}{0.08} = 2.95 \times 10^{-4} \text{ N}$$

3. [MJC/H2/Prelims 2010/P2/2 (part)]

- a. The average height of an adult is about 1.75 m.

$$\text{The potential difference between the head and feet: } \Delta V = Ed = (1.0 \times 10^2) \times (1.75) = 175 \text{ V}$$

[Acceptable range of average height of an adult: 1.50 m to 2.00 m]

- b. 175 V is a relatively high potential difference. It would have given a nasty electric shock if it were applied by the power mains. This is because the power mains has an infinite supply of electric charge in the forms of electrons. A large current would thus have conducted through the body.

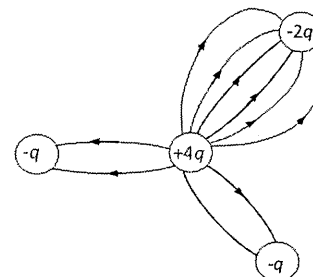
However, the number of free electrons in the atmosphere is very small. Thus, although there is a large potential difference between the head and feet, this potential difference is not able to deliver sufficient current through the body. Therefore, an electric shock is not experienced.

4. [NJC/H2/Prelims 2010/P2/3]

- a. Any two

- Field lines should not cross one another.
- Fields line should originate from and enter the charge perpendicularly.
- The number of field lines interacting with the $-2q$ charge by the $+4q$ charge should be more than the lines interacting with the $-q$ charge.
- Straight, parallel field lines are used for uniform electric field. The field lines interacting by the $+4q$ should not be parallel as it is a non-uniform electric field.
- The shape of the electric field lines interacting 2 charges of different magnitude are asymmetrical.

- b.



- c. Total electric potential at the $+4q$ charge:

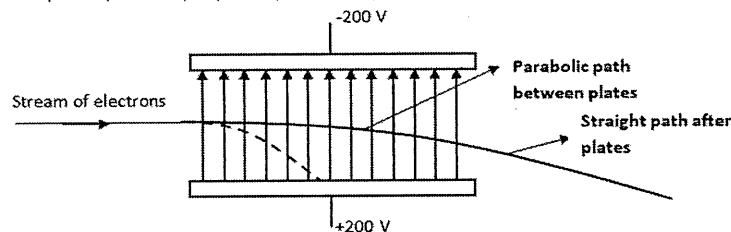
$$V = \frac{1}{4\pi\epsilon_0} \left(\frac{-q}{x} + \frac{-q}{x} + \frac{-2q}{x} \right) = \frac{1}{4\pi\epsilon_0} \frac{-4q}{x}$$

Work done by an external agent to bring the $+4q$ charge to infinity:

$$W = Q\Delta V = (+4q) \times \left(0 - \frac{1}{4\pi\epsilon_0} \frac{-4q}{x} \right) = \frac{16q^2}{4\pi\epsilon_0 x}$$

5. [NYJC/H2/Prelims 2010/P2/5]

- a. The electric field strength at a point is defined as the electric force exerted per unit positive charge placed at that point.
- b. i. Electric field lines are parallel and equally spaced.
Either path 1 (solid line) or path 2 (broken line) is accepted.



$$\text{ii. } E = \frac{V}{d} \Rightarrow d = \frac{V}{E} = \frac{400}{2.0 \times 10^4} = 0.0200 \text{ m}$$

- iii. Take upwards to be positive (i.e. upward electric field is positive).

$$a_y = \frac{F_y}{m_e} = \frac{qE_y}{m_e} = \frac{(-1.6 \times 10^{-19}) \times (2.0 \times 10^4)}{9.11 \times 10^{-31}} = -3.51 \times 10^{15} \text{ m s}^{-2}$$

- iv. Time taken for the electron to travel through the entire length of the plate:

$$t = \frac{s_x}{v_x} = \frac{0.040}{1.0 \times 10^8} = 4.00 \times 10^{-10} \text{ s}$$

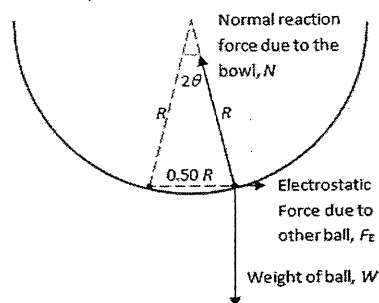
In the vertical direction, take upwards to be positive.

$$s_y = u_y t + \frac{1}{2} a_y t^2 = 0 + \frac{1}{2} (-3.5126 \times 10^{15}) (4.0 \times 10^{-10})^2 = -2.81 \times 10^{-4} \text{ m}$$

6. [RI/H2/Prelims 2010/P2/5]

a. i. $\sin \theta = \frac{0.25R}{R} \Rightarrow \theta = 14.5^\circ$ (shown)

- ii. All forces present in their correct directions



iii. $N \sin \theta = F_e$

$$N \cos \theta = mg$$

$$\Rightarrow F_e = mg \tan \theta$$

$$\Rightarrow \frac{1}{4\pi\epsilon_0} \frac{(15 \times 10^{-9} \text{ C})^2}{(0.5R)^2} = (0.5 \times 10^{-3})(9.81) \tan(14.5^\circ)$$

$$\Rightarrow R = 0.0799 \text{ m}$$

- b. θ will be smaller.

7. [RVHS/H2/Prelims 2010/P2/4]

- a. Coulomb's Law is only applicable to point charges.

In order for the formula to be valid, the distance $d \gg$ radius of the conducting spheres.

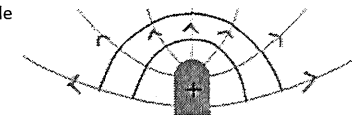
Or

Coulomb's law works for point charges or spherical charge distributions. Since the two spheres are conducting, the charges they contain will be repelled to the far sides and the charge distribution will no longer be spherical, which means that the formula will not work.

- b. i. A: constant; B: decreasing
- ii. The tangent of the field lines represents the direction of the force that a test positive charge experiences. Thus, if the field lines touched or crossed, the force a test positive charge experiences at that point would not be unique which is not possible.

8. [TJC/H2/Prelims 2010/P3/2]

- a. i. Curve concave downwards, crossing all field lines at right angle



- ii. Electrons are stripped off from the molecules.

Electrons and positive ions move off in opposite directions, giving rise to an electric current.

b. i. 1. $E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \Rightarrow Q = 4\pi\epsilon_0 r^2 E = 4\pi \times (8.85 \times 10^{-12}) \times (2.00 \times 10^{-3})^2 \times (1.13 \times 10^7) = 5.03 \times 10^{-9} \text{ C}$

2. $V = \frac{1}{4\pi\epsilon_0} \frac{Q}{r} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \times r = E \times r = (1.13 \times 10^7) \times (2.00 \times 10^{-3}) = 2.26 \times 10^4 \text{ V}$

- ii. 1. Electrostatic force experienced by the lower sphere due to the upper sphere:

$$F_e = \frac{1}{4\pi\epsilon_0} \frac{Q_1 Q_2}{r^2} = \frac{1}{4\pi \times (8.85 \times 10^{-12})} \frac{(5.03 \times 10^{-9})^2}{(4.0 \times 10^{-2})^2} = 1.42 \times 10^{-4} \text{ N}$$

Consider vertical equilibrium of the lower sphere:

$$N = F_e + mg$$

$$m' = \frac{N}{g} = \frac{F_e + mg}{g} = \frac{1.422 \times 10^{-4}}{9.81} + 8.205 \times 10^{-3} = 1.449 \times 10^{-5} + 8.205 \times 10^{-3} = 8.219 \times 10^{-3} \text{ kg}$$

Hence, the balance shows the reading of 8.219 g.

2. Due to repulsion, more charges accumulate on the outer sides of the 2 spheres. Thus, the distance between the centres of charge of the spheres increases.

The electrostatic force F will be smaller and so the reading m' will be less. Thus the answer in b.ii.1 is an over-estimation.

9. [YJC/H2/Prelims 2010/P2/1]

- a. See definitions in lecture notes.
- b. i. Electric potential energy
- ii. Assuming that the uranium nucleus is much more massive than the proton / the uranium nucleus remains at rest, and that initially the proton is sufficiently far from the uranium nucleus such that its initial electric potential energy is virtually zero.

By conservation of energy,

$$E_{p,i} + E_{k,i} = E_{p,f} + E_{k,f}$$

$$0 + \frac{1}{2}mv_i^2 = \frac{1}{4\pi\epsilon_0} \frac{q_1q_2}{r} + 0$$

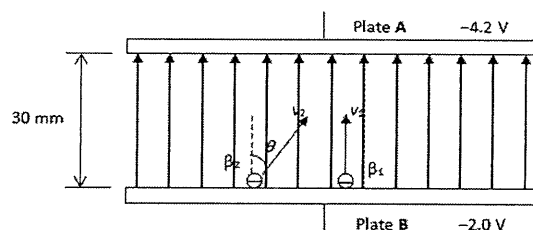
$$\frac{1}{2}(1.67 \times 10^{-27})v^2 = \frac{1}{4\pi \times (8.85 \times 10^{-12})} \frac{(1.60 \times 10^{-19})[92 \times (1.60 \times 10^{-19})]}{1.5 \times 10^{-14}}$$

$$v = 4.11 \times 10^7 \text{ m s}^{-1}$$

- iii. The alternative particle would be the neutron.
The neutron has no charge; thus it can penetrate the nucleus easily without being repelled.

10. [YJC/H2/Prelims 2010/P3/7]

- a. i. Electric field strength at a point is the electric force exerted per unit positive charge placed at that point.
The SI unit is N C^{-1} or V m^{-1} .
- ii. Electric field strength = $\Delta V / d$
- b. i. Parallel and equally spaced arrows



ii. $E = \frac{\Delta V}{d} = \frac{(-2.0 \text{ V}) - (-4.2)}{0.030 \text{ m}} = 73.3 \text{ N C}^{-1}$

- iii. By energy conservation, loss in KE = gain in EPE,

$$\Rightarrow \frac{1}{2}mv_1^2 - 0 = e\Delta V$$

$$\Rightarrow v_1 = \sqrt{\frac{2e\Delta V}{m}} = \sqrt{\frac{2 \times (1.60 \times 10^{-19})[(-2.0) - (-4.2)]}{9.11 \times 10^{-31}}} = 8.79 \times 10^5 \text{ m s}^{-1}$$

- iv. 1. Even though the distance has been halved, as long as the p.d. remains the same, the velocity v_1 will remain the same because the p.d. does not depend on the distance.
2. As plate A is now at a higher potential than plate B, the electron emitted from plate B will always be attracted to plate A. Thus, the minimum velocity required will be zero.

- v. By energy conservation, loss in KE = gain in EPE,

$$\frac{1}{2}mv_2^2 - \frac{1}{2}m(v_1 \sin \theta)^2 = e\Delta V$$

$$\frac{1}{2}mv_2^2(1 - \sin^2 \theta) = e\Delta V$$

$$\cos \theta = \sqrt{\frac{2e\Delta V}{mv_2^2}} = \sqrt{\frac{2 \times (1.60 \times 10^{-19})[(-2.0) - (-4.2)]}{(9.11 \times 10^{-31}) \times (1.1 \times 10^6)^2}} = 0.79916$$

$$\theta = 36.9^\circ$$

- c. i. Negative charge

ii. At vertical equilibrium, $F_g = F_e \Rightarrow mg = qE \Rightarrow \frac{q}{m} = \frac{g}{E} = \frac{9.81}{9.67} = 1.01 \text{ C kg}^{-1}$

iii. If the mass of the particle is $72 \times 10^{-19} \text{ kg}$, its charge is:
 $q = (1.0145) \times (72 \times 10^{-19}) = 7.3042 \times 10^{-18} \text{ C} = 5.651 \times (1.60 \times 10^{-19} \text{ C}) = 45.651e$

This is not valid as charge is quantised in integer multiples of e .